
Lecture 9

ECE 4107 (Section A)

Telecommunication Engineering



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STORED PROGRAM CONTROL

- Program or set of instructions to the computer are stored in its memory.
- Instructions executed automatically one by one by the processor.
- Programs are for Telephone exchange switching control functions.
- Hence called SPC.



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FEATURES

- ❑ Full scale automation of exchange functions.
- ❑ Common channel signaling
- ❑ Centralized maintenance
- ❑ Automatic fault diagnosis.
- ❑ Interactive human machine interface.

- In telecommunication, common-channel signaling, or common-channel interoffice signaling, is the transmission of control information via a separate channel



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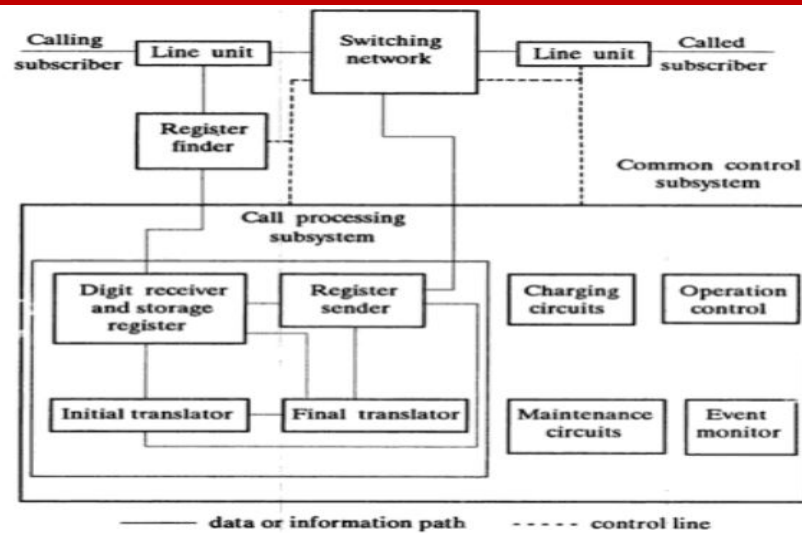
- ❑ **REQUIREMENTS FROM COMPUTER-**
- ❑ Telephone exchange must operate without interruption 24 hrs a day, 365 days a year. for years to come.



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Common Control Switching System



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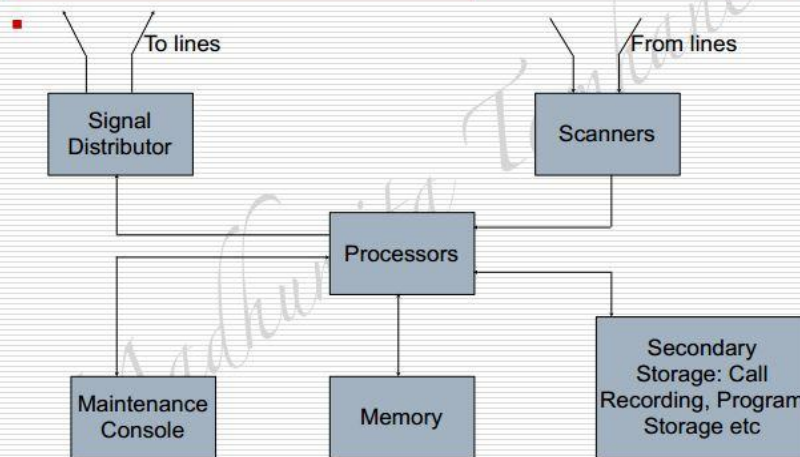
Centralised SPC

- ❑ Control equipments must be replaced by a single powerful computer.
- ❑ Must be capable of processing more than 100 calls per second along with other tasks.
- ❑ May use more than one processor for redundancy.
- ❑ Each processor has access to all exchange resources and function programs.
- ❑ Each processor capable of executing all control functions.

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Centralised SPC - no redundancy



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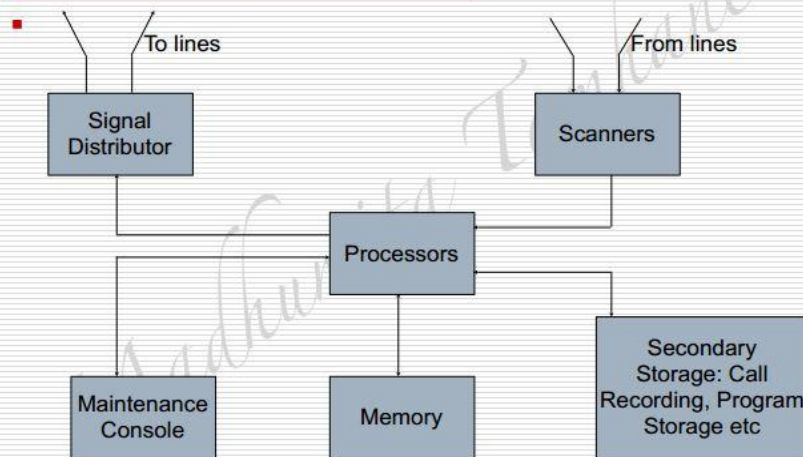
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- Practically, resources and memory modules are **shared** by processors.
- Each processors may have dedicated path to exchange resources.
- Each processors may have its own copy of programs and data in dedicated memory modules.
- Two Processor configuration is most common.

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Centralised SPC - no redundancy



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Modes of Dual Processor Architecture

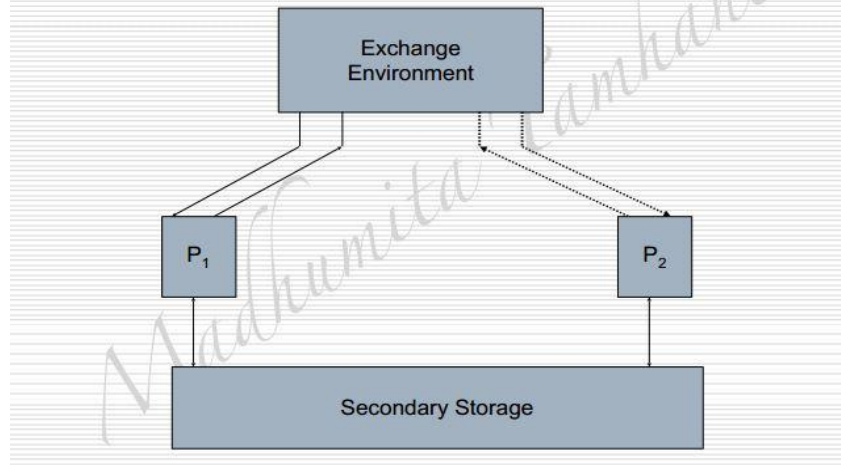
- Standby mode
- Synchronous mode
- Load sharing mode



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Standby mode



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Standby Mode

- Simplest
- H/W and S/W of one processor are active.
- Other is standby.
- Standby processor is brought to line only when active processor fails.
- Standby processor should be able to reconstitute the state of exchange system during takeover.
 - Which subscriber or trunk are busy or free.
 - Which paths connected through the network.



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Standby Mode

□ Small exchanges:

- By scanning all status signals during takeover.
- Only the calls being established at time of failure are disturbed.

□ Large exchanges:

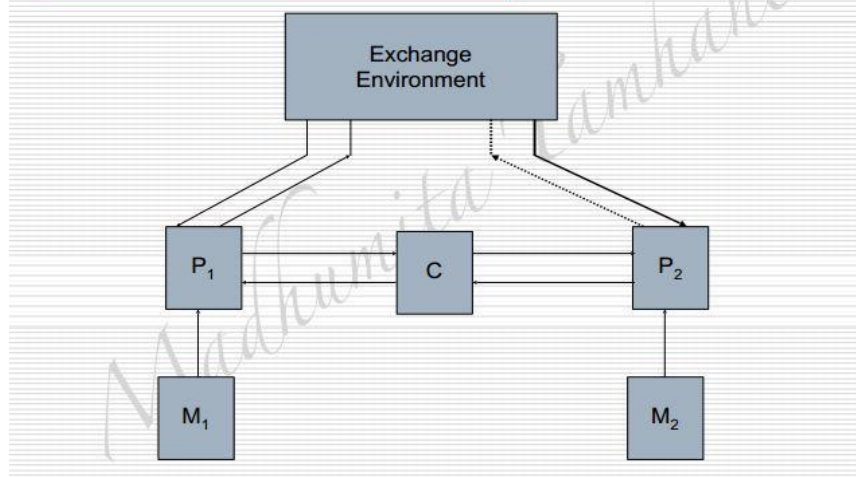
- Not possible to scan all status signals within reasonable time.
- Active processor copies the status of the system periodically into secondary storage.
- Most recent updates are taken by standby at takeover.
- All calls which changed status after last updates are disturbed.



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Synchronous duplex mode



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Synchronous duplex mode

- ❑ Both processors execute same instructions.
- ❑ Results compared continuously.
- ❑ During fault, comparator results mismatch.
- ❑ Each processor have same data in its memory.
- ❑ Each receive same information from exchange.
- ❑ One processor actually controls .
- ❑ Other synchronises but does not participate.



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Synchronous duplex mode

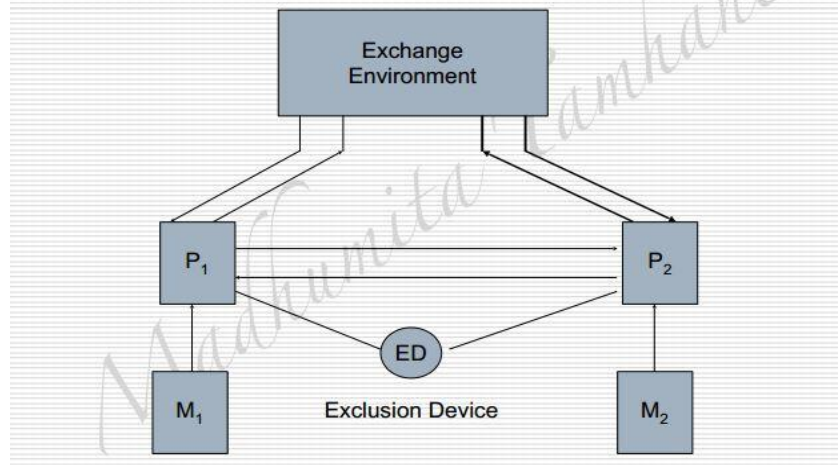
- ❑ During fault:
- ❑ P1 & P2 decoupled
- ❑ Run checkout program in each machine.
- ❑ Call processing suspended temporarily without disturbing the current call.
- ❑ Good processor takes control.
- ❑ Once repaired, other processor copies contents of active processor in its memory.
- ❑ Comparator is enabled.



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Load Sharing mode



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Load Sharing mode

- ❑ Both processors are active simultaneously.
- ❑ Both share the load and resources dynamically.
- ❑ Both processors have access to entire exchange.
- ❑ Incoming call is assigned randomly
- ❑ or in a predetermined order to one of the processors.
- ❑ Assigned processor handles the call through completion.
- ❑ Both have separate memories for storing temporary call data .

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Load Sharing mode

- Both are in mutual coordination through inter processor link.
- If information exchange fails, healthy processor takes over.
- Exclusion devise prevents both to be active together.
- Current calls are transferred.
- Calls being established are lost.



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Load Sharing mode

- Traffic sharing depends on the conditions of the processors and their requirements.
- During testing on one, other can take more traffic.
- Gives much better performance during traffic overloads.
- It's a step towards distribution control.



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Availability of the single processor system

- Main purpose of redundant configuration is to increase availability.
- Availability of single processor:
- $A = \text{MTBF} / (\text{MTBF} + \text{MTTR})$
- MTBF = mean time between failure
- MTTR = mean time to repair
- Unavailability $U = 1 - A$
- $= 1 - \{\text{MTBF} / (\text{MTBF} + \text{MTTR})\}$
- $\text{MTTR} / (\text{MTBF} + \text{MTTR})$
- If $\text{MTBF} \gg \text{MTTR}$
- $U = \text{MTTR} / \text{MTBF}$



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Availability of the Dual processor system

- A dual processor is said to have failed only when both the processors fail.
- System is totally unavailable.
- Condition - One processor has failed.
 - Other also fails before first is repaired.
- Conditional probability that second fails during MTTR of first.
- MTBF of dual processor can be given in terms of MTBF and MTTR of single processors as-
- $\text{MTBF}_D = (\text{MTBF})^2 / 2\text{MTTR}$ -using conditional probability.



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Availability of the Dual processor system

- $A_D = MTBF_D / (MTBF_D + MTTR)$
- $A_D = (MTBF)^2 / [(MTBF)^2 + 2(MTTR)^2]$
- $U_D = 1 - A_D$
- $= 2(MTTR)^2 / [(MTBF)^2 + 2(MTTR)^2]$
- If $MTBF \gg MTTR$
- $U_D = 2(MTTR)^2 / (MTBF)^2$



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- Given that $MTBF = 2000$ hours and $MTTR = 4$ hours, calculate the unavailability for single and dual processor system.

- $U = 4/2000 = 2 \times 10^{-3}$
- 525 hours in 30 years.
- $U_D = 2 \times 16/2000 = 8 \times 10^{-6}$
- 2.1 hours in 30 years

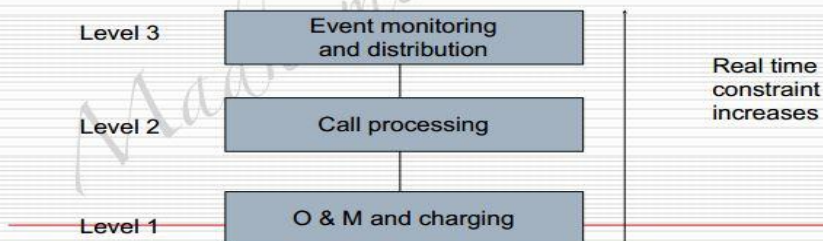


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Functions of control subsystem

- Event monitoring
- Call processing
- Charging
- Operation and maintenance
- Grouped under 3 levels



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Functions of control subsystem

- Event monitoring has highest priority, O&M and then charging the least.
- Real time constraint asks for priority interrupts.
- If an EVENT occurs during O&M, it will be interrupted and event will be handled.
- Then O&M will be resumed.
- Nesting interrupt to suspend low level functions and take up higher level functions.



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Functions of control subsystem

□ Interrupt processing

Level n process
Suspend level n
Take up level n + x
Suspend level n + x
Take up level n + x + y
Level n + x + y complete
Resume level n + x
Level n + x complete
Resume level n



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